

## One Haar-Random State Looks Like Many

*Classical outputs extracted from  $t$  copies of a single Haar-random state should be statistically indistinguishable from outputs extracted from  $t$  independent ones*

**Status.** Statement: AI-written from Amit Behera, Zvika Brakerski, Or Satath and Omri Shmueli, *Pseudorandomness with Proof of Destruction and Applications*, IACR ePrint 2023/543, TCC 2023. Not yet checked by a human. The source states this as a conjecture it believes true, says it could neither prove it nor find it formalised anywhere, and proves a strictly weaker pointwise domination bound (its Lemma 2) that suffices for its own applications.

**Category.** *Quantum information; quantum cryptography.*

**Conjecture (informal).** *Hand an algorithm one copy of a random quantum state and let it print a classical string. Do this  $t$  times. There are two ways to supply the states: draw  $t$  independent Haar-random states, or draw one Haar-random state and hand out  $t$  copies of it. In the second, the  $t$  printed strings are correlated — they all come from the same underlying state. The conjecture is that no one can tell: for any algorithm and any polynomial  $t$ , the two joint distributions of classical strings are within negligible total variation distance. It is an entirely information-theoretic statement, with no computational assumption anywhere, and the source could neither prove it nor find it in the literature.*

### 1 The setting

Let  $\mathcal{H}_n$  be the state space of  $n$  qubits, of dimension  $N = 2^n$ , and let  $\mu_{\mathcal{H}_n}$  be the Haar measure on its pure states. Let  $\mathcal{A}$  be any algorithm that takes a single  $n$ -qubit state and outputs a classical string; it may be an arbitrary quantum channel followed by a measurement, and it may use ancillae, so it need not be a projective measurement on the input register alone.

Two distributions over  $t$ -tuples of classical strings arise, and the conjecture compares them.

- The *product* distribution: draw  $|\phi_1\rangle, \dots, |\phi_t\rangle$  independently from  $\mu_{\mathcal{H}_n}$  and output  $\mathcal{A}(|\phi_1\rangle) \otimes \dots \otimes \mathcal{A}(|\phi_t\rangle)$ .
- The *correlated* distribution: draw a single  $|\phi\rangle \sim \mu_{\mathcal{H}_n}$ , and output  $\mathcal{A}(|\phi\rangle) \otimes \dots \otimes \mathcal{A}(|\phi\rangle)$ , where  $t$  independent invocations of  $\mathcal{A}$  each receive one copy of the same  $|\phi\rangle$ .

Why the two might differ is easy to see. If  $\mathcal{A}$  measures in the computational basis, the correlated experiment draws  $t$  samples from one Porter-Thomas distribution while the product experiment draws each sample from a different one; second moments already separate the two families, and the question is whether the separation is visible in total variation at polynomial  $t$ .

## 2 Notation and parameters

- $n$  is the number of qubits and  $N = 2^n$  the dimension; *negligible* is in  $n$ .
- $t = t(n)$  is any polynomial; the conjecture is not claimed for superpolynomial  $t$ , and cannot hold for  $t$  comparable to  $N$ .
- $\mathcal{A}$  ranges over *all* algorithms with classical output, not only efficient ones — the statement is information-theoretic.
- TV denotes total variation distance between the two distributions over  $t$ -tuples of classical strings.

## 3 The conjecture

**Conjecture 1** ([BBSS23, §1.3]). *For any algorithm  $\mathcal{A}$  that outputs a classical string, and any polynomial  $t = t(n)$ ,*

$$\text{TV}(\mathcal{A}(|\phi_1\rangle) \otimes \dots \otimes \mathcal{A}(|\phi_t\rangle), \mathcal{A}(|\phi\rangle) \otimes \dots \otimes \mathcal{A}(|\phi\rangle))$$

*is negligible in  $n$ , where  $|\phi_1\rangle, \dots, |\phi_t\rangle \sim \mu_{\mathcal{H}_n}$  are independent and the same  $|\phi\rangle \sim \mu_{\mathcal{H}_n}$  is used in all the invocations on the right.*

**Remark 1** (How the source states it). Conjecture 1 is the source’s own sentence, page 10: “In some of the applications, we required the following quantum information-theoretic conjecture regarding Haar random states that we believe is true: For any algorithm  $\mathcal{A}$  that outputs a classical string, and any polynomial  $t = t(n)$ , the total variation distance between  $\mathcal{A}(|\phi_1\rangle) \otimes \dots \otimes \mathcal{A}(|\phi_t\rangle)$ , where

$|\phi_1\rangle, \dots, |\phi_t\rangle \sim \mu_{\mathcal{H}_n}$  and  $\mathcal{A}(|\phi\rangle) \otimes \dots \otimes \mathcal{A}(|\phi\rangle)$ , where the same state  $|\phi\rangle \sim \mu_{\mathcal{H}_n}$  is used in all the algorithms, is negligible in  $n$ .”

And on its status: “We could not prove nor find any previous work in the literature that proves or even formalizes this conjecture.” The notation  $\mathcal{A}$ , TV and  $N$  above is this statement’s; the mathematics is the source’s, unchanged.

**Theorem 1 (What is proved instead, [BSS23, Lemma 2]).** *Let  $\mathcal{A}$  output  $c$ -bit strings. For every  $t \in \text{poly}(\lambda)$  and all  $a_1, \dots, a_t \in \{0, 1\}^c$ ,*

$$\Pr_{\text{corr}}[(f_1, \dots, f_t) = (a_1, \dots, a_t)] \leq \frac{N^t}{\binom{N+t-1}{t}} \cdot \Pr_{\text{prod}}[(f_1, \dots, f_t) = (a_1, \dots, a_t)],$$

where *corr* and *prod* are the correlated and product distributions of §1.

**Remark 2 (Why Theorem 1 is much weaker than Conjecture 1).**

For  $t \ll N$  the factor  $N^t / \binom{N+t-1}{t}$  is about  $t$  factorial, so the lemma says only that the correlated distribution never exceeds the product distribution by more than a factorial factor, pointwise. That is enough to transfer a bound of the form “this bad event has probability  $\eta$  under the product distribution” into “at most  $t$  factorial times  $\eta$  under the correlated one”, which is what the source’s unforgeability and collision-freeness arguments need. It says nothing at all about total variation distance, which the factorial factor swamps.

The source is explicit that the lemma is a stand-in: “We believe that the distributions are in fact, statistically close due to the strong concentration of the Haar measure, but we have not been able to prove it. The lemma is a weaker version of this statement, but it suffices for our purposes.”

The proof of Theorem 1 in the special case where  $\mathcal{A}$  is a computational-basis measurement is two lines of symmetric-subspace counting: for each outcome tuple there is a unique symmetric type contributing, giving probability at most  $1/\binom{N+t-1}{t}$  under the correlated distribution and exactly  $N^{-t}$  under the product one. That argument gives the ratio and, being pointwise, cannot give more.

**Remark 3 (Where the difficulty is).** Three features make this harder than it first looks.

*The algorithm is arbitrary.*  $\mathcal{A}$  need not be a projective measurement on the input register: it may use ancillae, so the overall map

on the input is not unitary. The source flags exactly this elsewhere — “the destruct algorithms may use ancillae qubits, and therefore the overall process becomes non-unitary” — and non-unitary processing does not preserve Haar-randomness of a residual state, which is what a reduction to the projective case would want.

*The output may be long.* For a single-bit output, concentration of measure settles it quickly. For  $c$ -bit outputs the correlated distribution has a genuinely larger collision probability — of order  $2/N$  against  $1/N$  for the product distribution — and any proof must show that all such second- and higher-order discrepancies stay below negligible in total variation once summed over the  $\binom{t}{2}$  pairs and beyond.

*The bound must be uniform in  $\mathcal{A}$ .* The claim quantifies over all algorithms with no efficiency restriction, so it is a statement about the  $t$ -copy Haar ensemble itself and not about what a bounded observer can do with it.

**Remark 4 (Why it is worth settling on its own).** The source’s own reason: “We think this is an interesting open question on its own, and if proven, this result can be a useful tool for quantum cryptography.” It is the kind of statement that gets reproved ad hoc, in a weaker form each time, wherever a security proof needs to move between one shared Haar-random state and many independent ones. A clean version with an explicit bound in  $t$  and  $n$  would replace those.

A refutation would be at least as interesting: it would exhibit an algorithm whose classical output distinguishes “one state,  $t$  copies” from “ $t$  states” at polynomial  $t$ , which is a statement about the  $t$ -copy Haar ensemble that no one currently expects.

## 4 Bibliography

[BBS23] Amit Behera, Zvika Brakerski, Or Sattath and Omri Shmueli. *Pseudorandomness with proof of destruction and applications*. IACR ePrint 2023/543; TCC 2023. The source. The conjecture and its status are on page 10, in the Open Problems section; the correlated and product distributions are Definition 3; the pointwise bound is Lemma 2 with its proof sketch and footnote on page 18, and the general case in Appendix B.

- [JLS18] Zhengfeng Ji, Yi-Kai Liu and Fang Song. *Pseudorandom quantum states*. CRYPTO 2018, Part III, LNCS 10993, pages 126–152, Springer, 2018. The paper that started the study of pseudorandom states, and the source of the earlier Haar-indistinguishability conjecture in this line.
- [BS20b] Zvika Brakerski and Omri Shmueli. *Scalable pseudorandom quantum states*. CRYPTO 2020, Part II, LNCS 12171, pages 417–440, Springer, 2020. Cited by the source for scalable pseudorandom states; the same authors’ work in this line is where statistical closeness to the Haar ensemble is proved for a particular efficiently generated family.
- [AQY21] Prabhanjan Ananth, Luowen Qian and Henry Yuen. *Cryptography from pseudorandom quantum states*. arXiv:2112.10020, 2021. The post-selection construction of short-input function-like states from pseudorandom states that the source discusses as not obviously commuting with a destruct algorithm.
- [AGQY22] Prabhanjan Ananth, Aditya Gulati, Luowen Qian and Henry Yuen. *Pseudorandom (function-like) quantum state generators: new definitions and applications*. arXiv:2211.01444, 2022. Cited by the source for quantum adaptive pseudorandomness, one of the neighbouring open questions in the same list.